

### **Sources Sought Notice**

Department of the Interior

U.S. Geological Survey (USGS)

Geology, Minerals, Energy, & Geophysics Science Center (GMEGSC)

The U.S. Geological Survey (USGS) is conducting market research to determine the availability of qualified businesses capable of providing stable isotope analysis services, as described below.

This Sources Sought Notice is for information and planning purposes only and does not constitute a solicitation. This is not a Request for Quote (RFQ). The Government is not obligated to issue a solicitation or award a contract as a result of this notice. The Government will not reimburse any costs associated with responding.

The applicable North American Industry Classification System (NAICS) code is **541380 – Testing Laboratories**, with a size standard of \$19.0 million. The applicable Product Service Code (PSC) is **B529. SPECIAL STUDIES/ANALYSIS- SCIENTIFIC DATA**

All responsible sources may submit a capability statement demonstrating their ability to meet the requirement. Responses shall be submitted via email to Joel Berberena at [joel\\_berberena@ios.doi.gov](mailto:joel_berberena@ios.doi.gov) and must be received before the close date of this notice. Late responses or responses submitted by other than email will not be considered.

### **Capability Statement Requirements:**

Interested vendors shall provide the following:

1. Description of laboratory capabilities for stable isotope analysis ( $\delta^{13}\text{C}$  and  $\delta^{18}\text{O}$ ) of calcite samples
2. Description of instrumentation and analytical methods used
3. Experience performing similar isotope or geochemical analysis
4. Examples of similar work performed
5. Description of quality control or analytical precision practices
6. Company information, including business size, UEI, and SAM registration status

The Government is seeking information on vendors capable of meeting these requirements. Responses will be used to determine the appropriate acquisition strategy.

# **Statement of Work**

## **Stable Isotope Analysis of Foraminifera Samples**

**Requisition Date: 31 March 2026**

### **I. GENERAL INFORMATION**

#### **A. Introduction**

The U.S. Geological Survey (USGS), Pacific Abrupt Climate Events (PACE) Project within the Geology, Minerals, Energy, & Geophysics Science Center (GMEGSC), requires stable isotope ( $\delta^{13}\text{C}$  and  $\delta^{18}\text{O}$ ) analysis of calcite samples (foraminifera). These analyses support reconstruction of past ocean conditions, including temperature, salinity, and global ice volume.

#### **B. Background**

The objective of this effort is to develop oxygen and carbon isotope records from planktic and benthic foraminifera in marine sediment cores collected off the coast of California. These records provide stratigraphic context for sediment cores and support development of age models beyond the range of radiocarbon dating (>40,000 years).

The resulting data will be used to evaluate changes in oceanographic conditions during glacial and interglacial periods, including variability in the California Current and associated upwelling dynamics.

#### **C. Definitions**

No special definitions are required for successful performance of this requirement.

#### **D. Scope**

The Contractor shall perform stable isotope analysis of approximately 400 foraminifera calcite samples (benthic and planktic) to generate data supporting reconstruction of past climate conditions along the California coast.

The work includes measurement of isotopic ratios and reporting of results sufficient to support scientific analysis by the Government.

### **II. WORK REQUIREMENTS**

#### **A. Technical Requirements**

1. The Government will provide crushed foraminifera samples. Each sample will be within a weight range of approximately 0.050–0.100 mg.
2. The Contractor shall perform stable isotope analysis using appropriate laboratory instrumentation capable of measuring  $\delta^{13}\text{C}$  and  $\delta^{18}\text{O}$  in calcite samples.
3. The Contractor shall measure:
  - a. Stable isotopic ratios of  $\delta^{18}\text{O}$  and  $\delta^{13}\text{C}$  for foraminifera calcite samples
  - b. International calcite standards
  - c. Internal precision associated with the analytical measurements

4. The Contractor shall compile analytical results, including measured values and associated analytical error, for all samples and standards.

#### **B. Deliverables**

1. The Contractor shall provide an electronic report containing:
  - a. Stable isotope values ( $\delta^{13}\text{C}$  and  $\delta^{18}\text{O}$ ) for all samples analyzed
  - b. Results from international standards
  - c. Analytical error and/or precision information associated with the measurements
2. Deliverables shall be provided electronically in a format accessible to the Government.
3. The final deliverable shall be submitted no later than the end of the Period of Performance.

### **III. PERFORMANCE STANDARDS AND ACCEPTANCE**

#### **A. Performance Standards**

Performance will be considered acceptable when the Contractor:

1. Completes analysis of all samples provided
2. Provides required isotopic measurements and associated information as specified in Section II
3. Submits a complete electronic report containing all required data elements

#### **B. Acceptance**

The Government will review submitted deliverables to verify that all required data elements are included. Acceptance will be based on completeness of the submission and inclusion of the required analytical results and associated information.

### **IV. SUPPORTING INFORMATION**

#### **A. Place of Performance**

The work shall be performed at the awarded Contractor's laboratory.

#### **B. Period of Performance**

Reference SF18

#### **C. Government Furnished Property (GFP)**

The Government will furnish foraminifera samples for analysis. No other Government-furnished property will be provided.

#### **D. Special Considerations**

None.